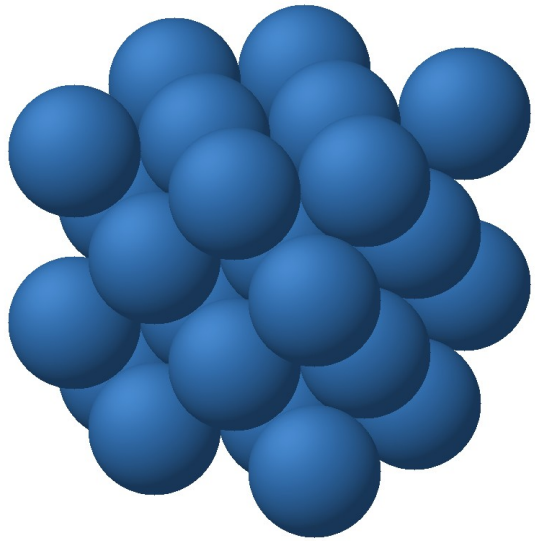
Interface Force Field

From atoms to evidence. A beginner’s guide to molecular mechanics and material interfaces.



Ideal fcc Rh. Sphere radius = half the nearest-neighbour distance.

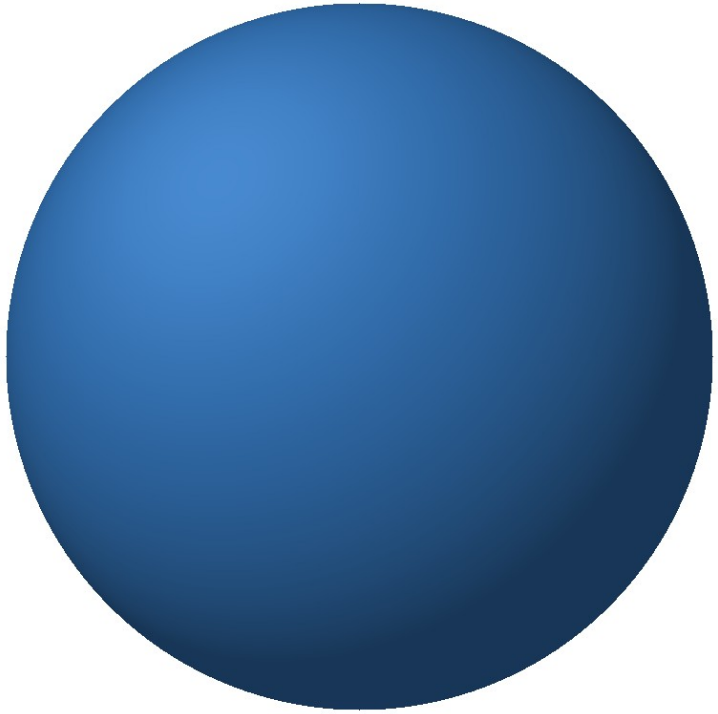
Sean Florez

Heinz et al., Langmuir (2013) · Kanhaiya et al., npj Comput. Mater. (2021)

An atom has a nucleus and electrons

The electrons help determine how atoms interact.

- The nucleus contains protons and neutrons.
- Electrons occupy a region around it.
- A sphere is a useful drawing of size and position.



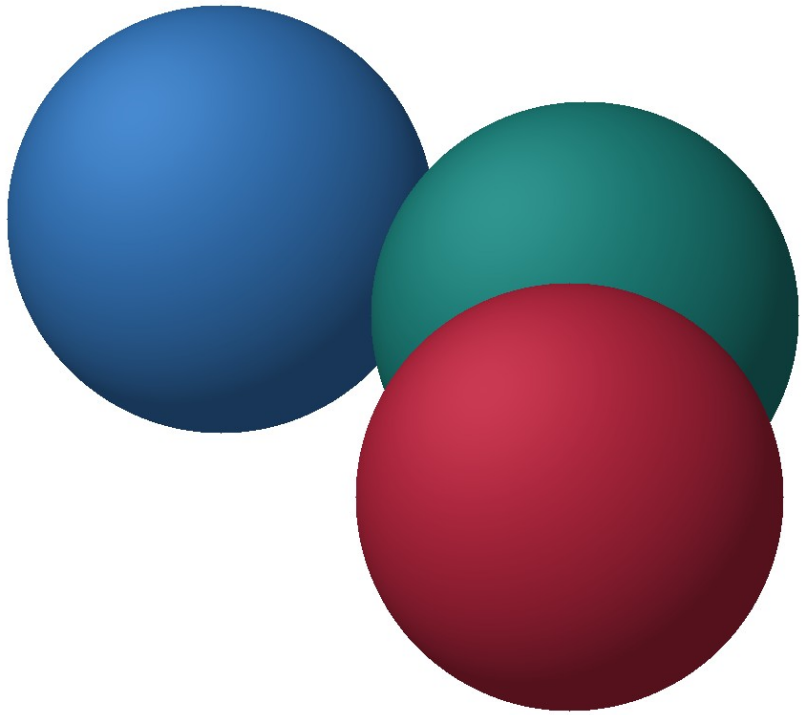
One sphere stands in for an atom and its electron cloud.

Original calculation or illustration

Molecules connect atoms; crystals repeat a pattern

The arrangement depends on the material.

- A molecule contains atoms joined by chemical bonds.
- A crystal repeats an arrangement through space.
- Water and a metal surface can share one simulation box.



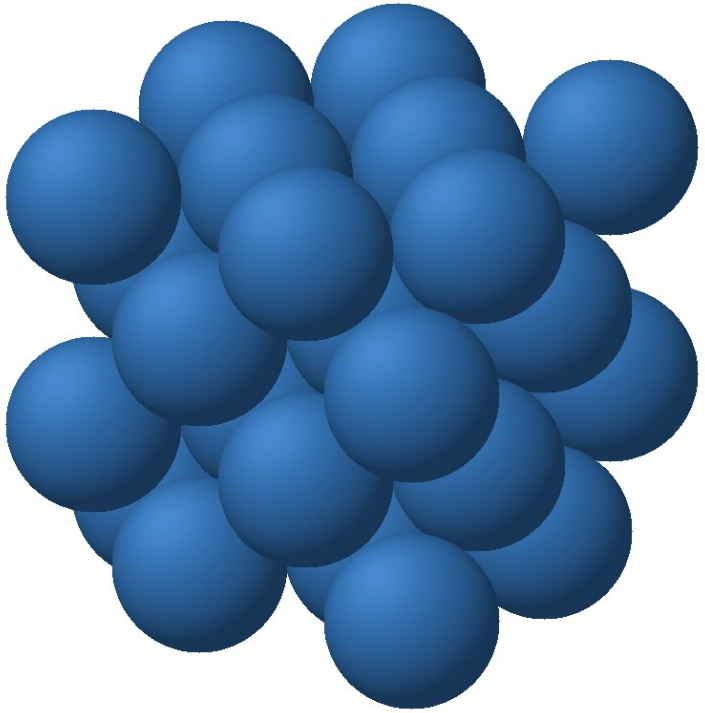
A connected group of atoms. Sphere sizes here are schematic.

Heinz et al., Langmuir (2013)

Atoms push and pull on one another

Changing their arrangement changes the material's energy.

- Atoms pushed too close resist further compression.
- At suitable separations, attraction can hold a solid together.
- The surrounding neighbors affect the balance.



Ideal fcc Rh. Sphere radius = half the nearest-neighbour distance.

Can a short molecule change a metal particle’s shape?

A published platinum study makes the question concrete.

System

Short chains of amino acids, called peptides, surround growing platinum crystals.

Change

Replacing one amino acid changes which crystal face the peptide favors.

Observation

Electron microscopy shows different particle shapes after synthesis.

Explanation

Simulations compare peptide binding on the competing faces.

Ruan et al., Nano Lett. (2013), doi:10.1021/nl400022g

A simulation is a controlled model experiment

Choose the question, construct a model, and calculate an observable.

- Inputs specify atoms, conditions and interactions.
- The calculation produces structures, motion or energies.
- A measured quantity tests the result under matched conditions.

Choose a question

Define a material and an observable.

Run a model

Compute interactions under stated conditions.

Check evidence

Compare predictions with measurements.

A model experiment answers a specific question under specific assumptions.

Different methods keep different physical detail

Choose the detail needed to answer the question.

- Density functional theory (DFT) describes electrons and their energy.
- Classical models use effective interactions between atoms.
- Learned potentials infer energies and forces from reference data.

Quantum / DFT

Describe electrons.
Small systems or
short time spans.

Classical / IFF

Use fitted interactions.
More atoms and
longer sampling.

Learned potential

Learn reference energies.
Check training coverage
and sampling.

Cost and accuracy depend on the system, software, hardware and reference.

Hohenberg & Kohn, Phys. Rev. (1964) · Heinz et al., Langmuir (2013) · Batatia et al., MACE (2022)

Compare the scale and the work actually timed

A single energy evaluation and a trajectory measure different work.

Calculation	Atoms	Work or simulated time	Reported cost
DFT, Trp-cage	281	One energy evaluation	21 min
AI²BMD, Trp-cage	281	One energy evaluation	0.072 s
Classical metal lattice	500	100 ps + 500 ps averaging	No matched timing

A picosecond (ps) is one trillionth of a second. AI²BMD timing used an NVIDIA A6000 and 32 CPU cores; costs depend on the benchmark.

Molecular mechanics assigns energy to atom positions

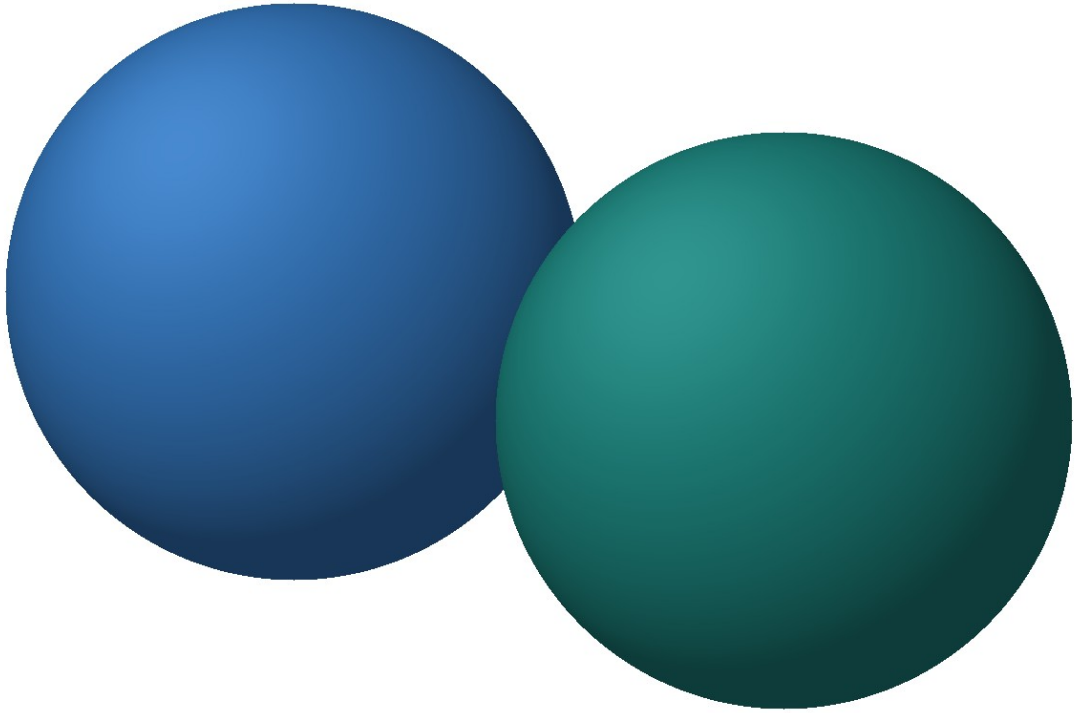
An energy function turns a configuration into a number.

- Potential energy describes the stored energy of an arrangement.
- The force field contains the equation, atom types and numerical parameters.
- Forces follow from how the energy changes when atoms move.
- Different physical effects contribute to the total energy.

Stretching a bond costs energy

Near its preferred length, a bond behaves like an effective spring.

- A chemical bond connects neighboring atoms.
- Compression or stretching raises the spring energy.
- The spring constant controls the local stiffness.



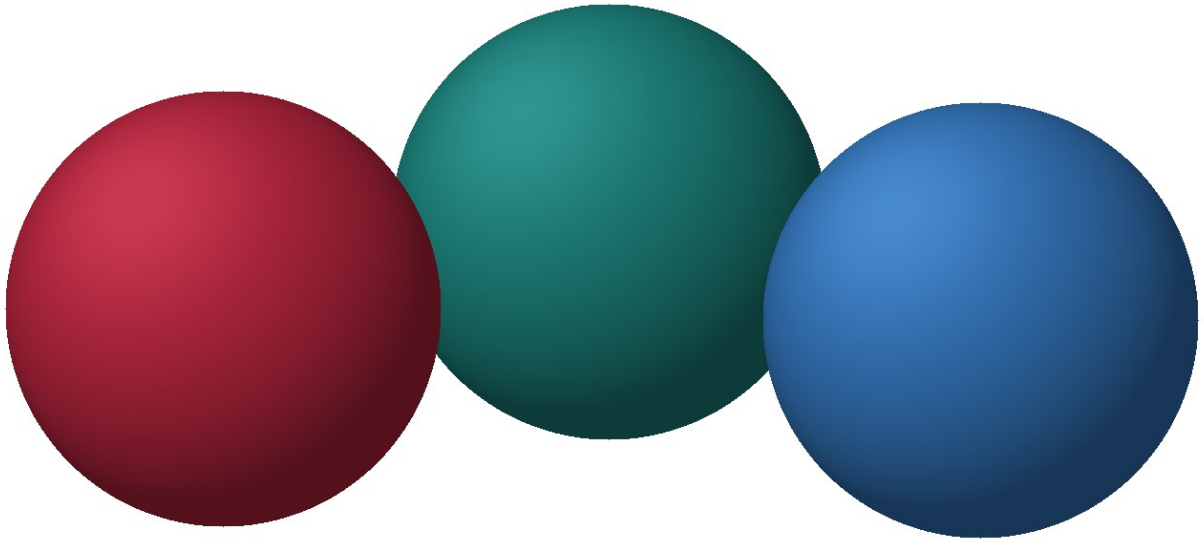
Preferred separation. Overlap or separation costs potential energy.

Heinz et al., Langmuir (2013) · Original calculation or illustration

Bending an angle also costs energy

Three connected atoms define a bond angle.

- A preferred angle helps set molecular shape.
- Changing that angle raises the bending energy.
- Bond lengths and angles describe different motions.

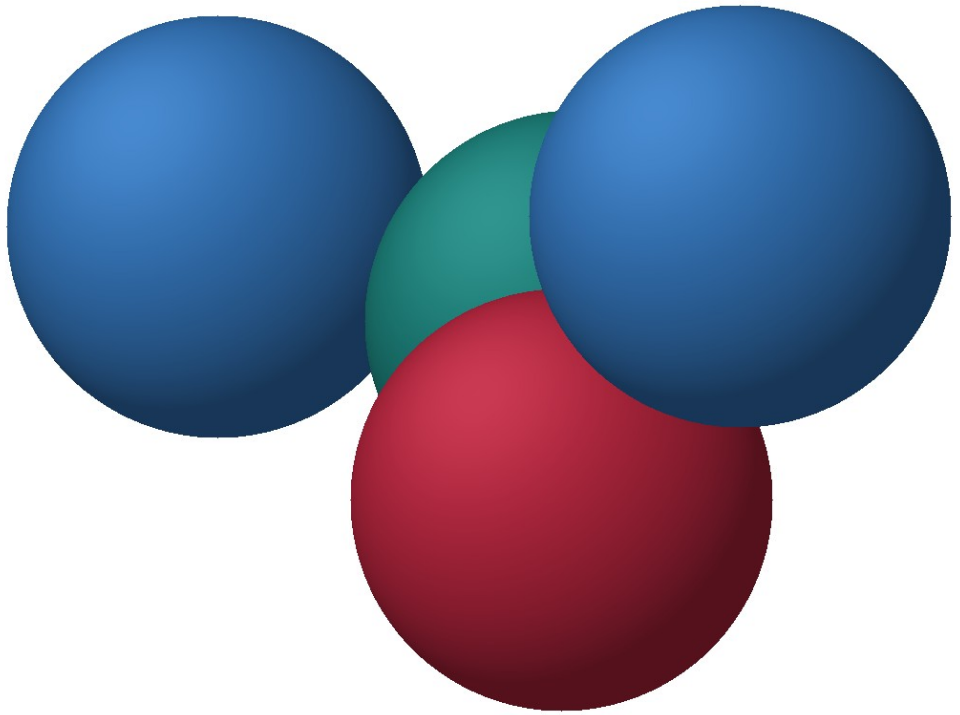


The angle is defined by three atom centres. Adjacent spheres touch.

A torsion describes a twist along a bond

Connected groups can prefer some orientations over others.

- Four connected atoms define a dihedral angle.
- Rotation changes the relative positions of nearby groups.
- A torsion term can have several preferred orientations.

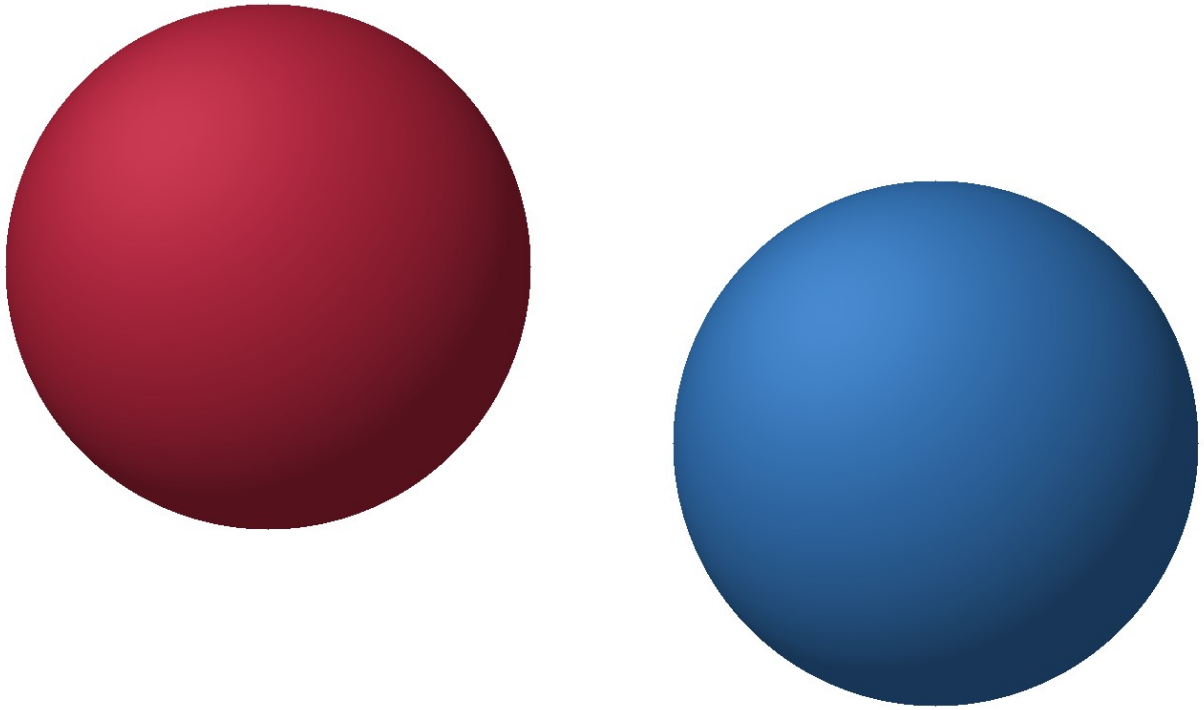


Four atom centres define a twist around the middle connection.

Atoms interact across a gap, too

Nonbonded terms describe interactions beyond a chemical bond.

- Van der Waals terms combine short-range repulsion and attraction.
- Coulomb terms describe interactions between model charges.
- Water competes with a molecule for contacts at a surface.

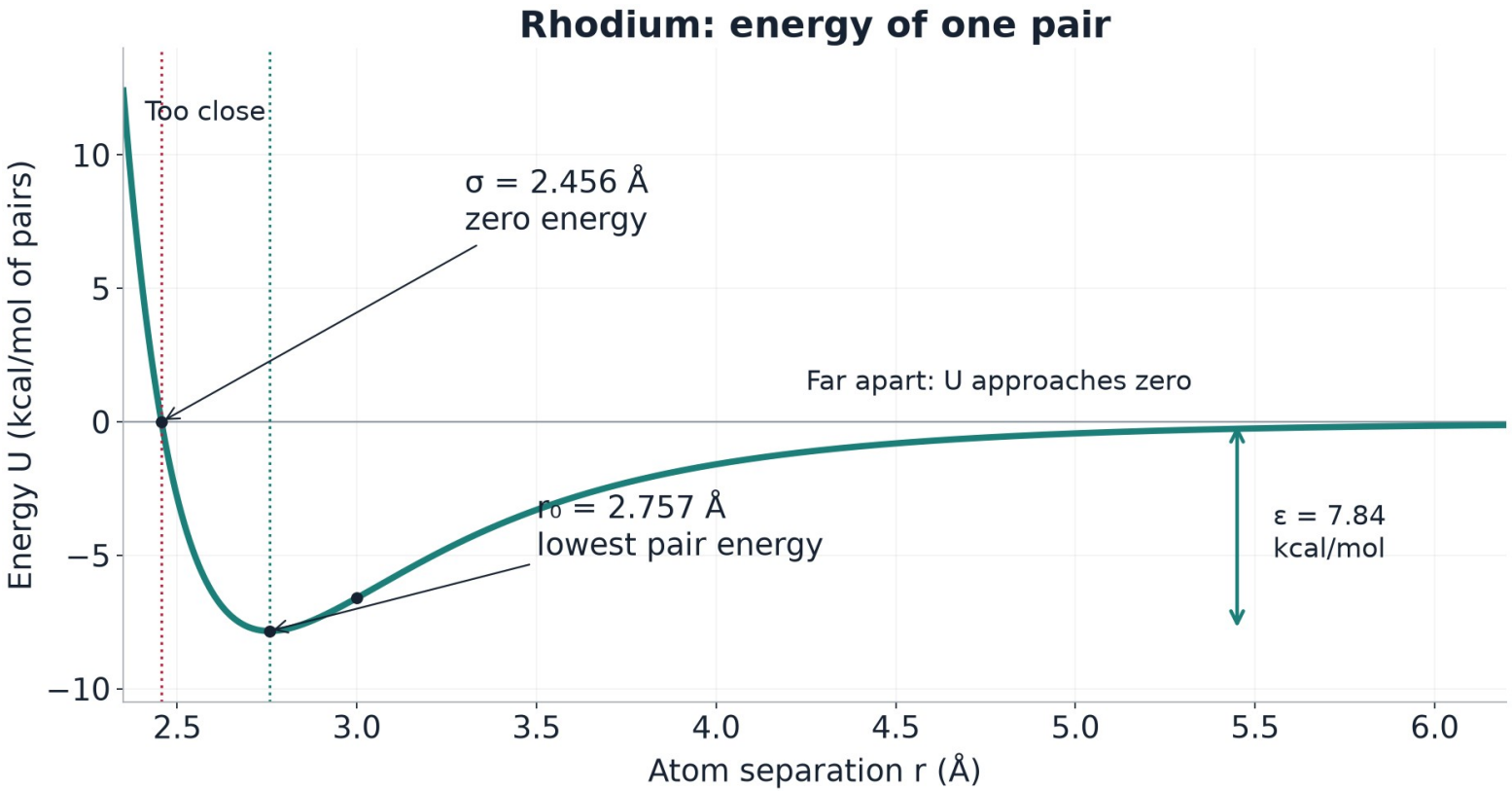


Space-filling atom models can interact across a gap without a chemical bond.

The Lennard-Jones curve has a wall and a well

This is the published rhodium 12-6 pair model.

- At short distance the energy rises steeply.
- The attractive well lies below the separated-pair reference.
- At large distance this pair energy approaches zero.

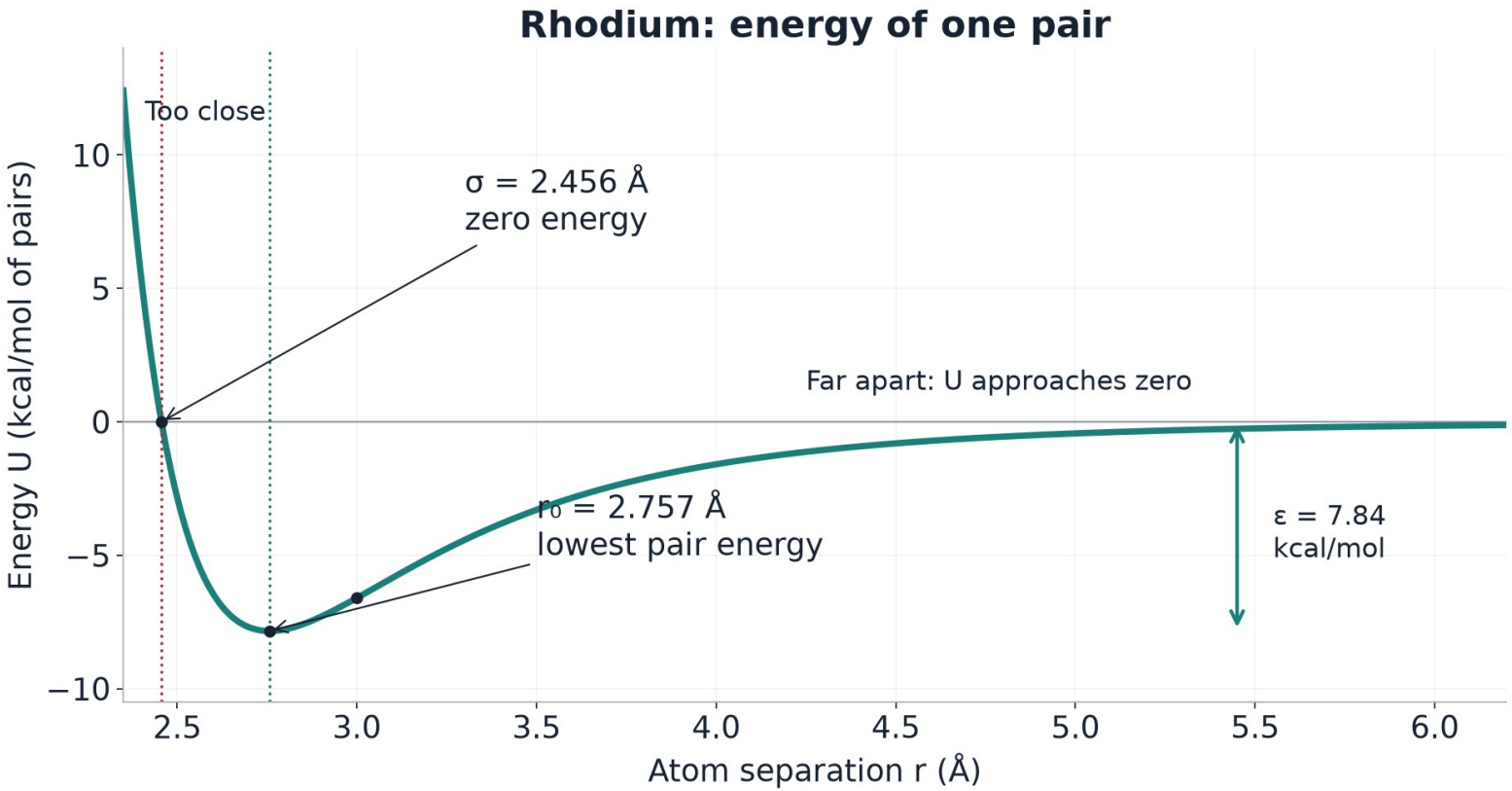


Analytical illustration from Kanhaiya et al. (2021), Eq. 1 and Table 1. Not a trajectory.

Read the zero crossing, minimum and well depth

An ångström (Å) is 0.1 nm. Here, kcal/mol reports energy per mole of pairs.

- r_0 = 2.757 Å marks the pair minimum.
- ε = 7.84 kcal/mol sets its depth.
- The textbook zero-crossing σ is 2.4562 Å.

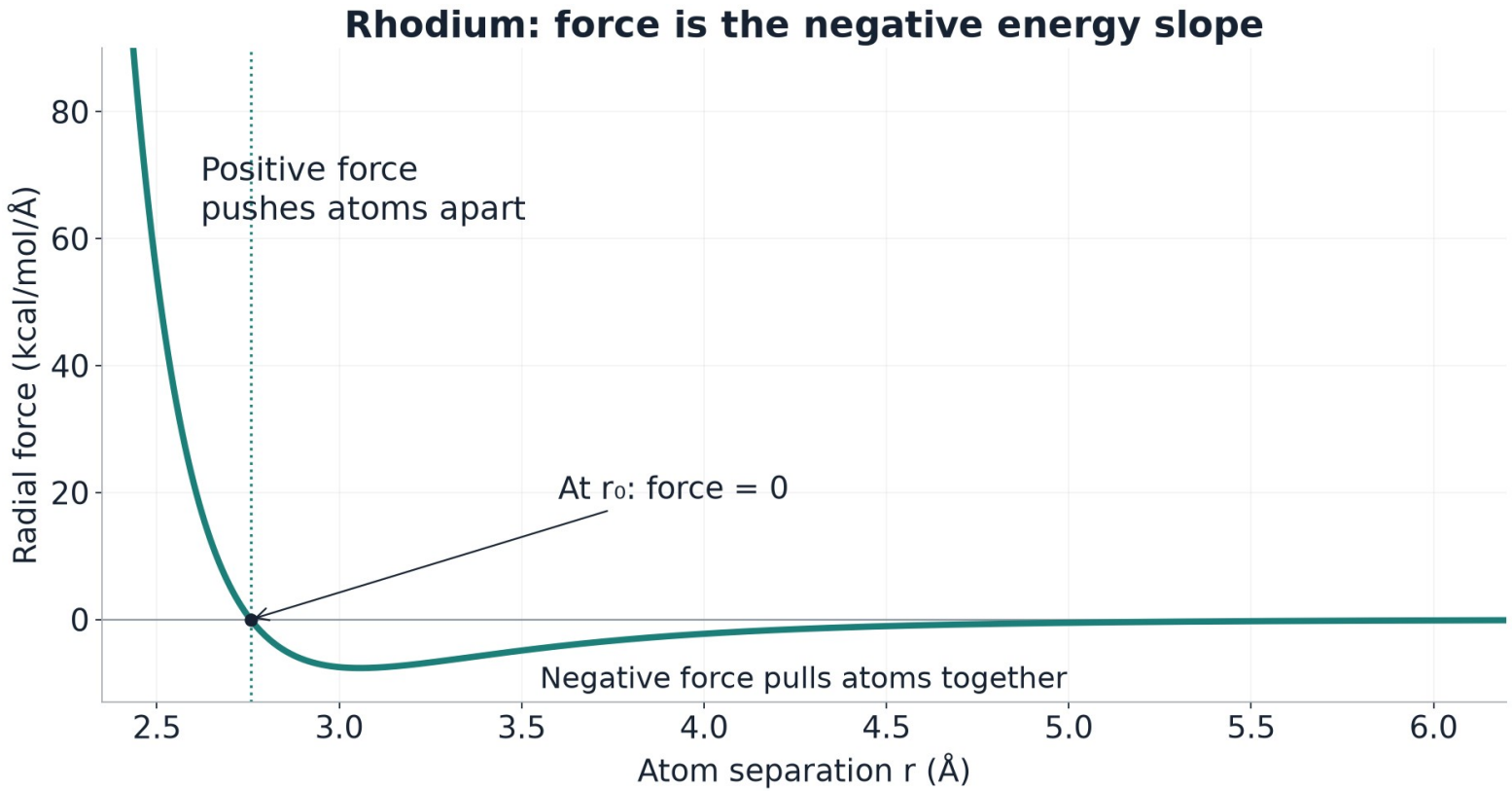


Analytical illustration from Kanhaiya et al. (2021), Eq. 1 and Table 1. Not a trajectory.

Force is the negative slope of energy

The direction depends on the slope at the current distance.

- Positive radial force pushes atoms apart.
- Negative radial force draws atoms together.
- The force is zero at the pair minimum.



Analytical illustration from Kanhaiya et al. (2021), Eq. 1 and Table 1. Not a trajectory.

Work the rhodium numbers

Substitute a distance into the energy and force equations.

Distance, Å	Energy, kcal/mol	Force, kcal/(mol Å)	Direction
2.500	−2.838	+54.072	Repulsion
2.757	−7.840	0	Pair equilibrium
3.000	−6.601	−7.511	Attraction

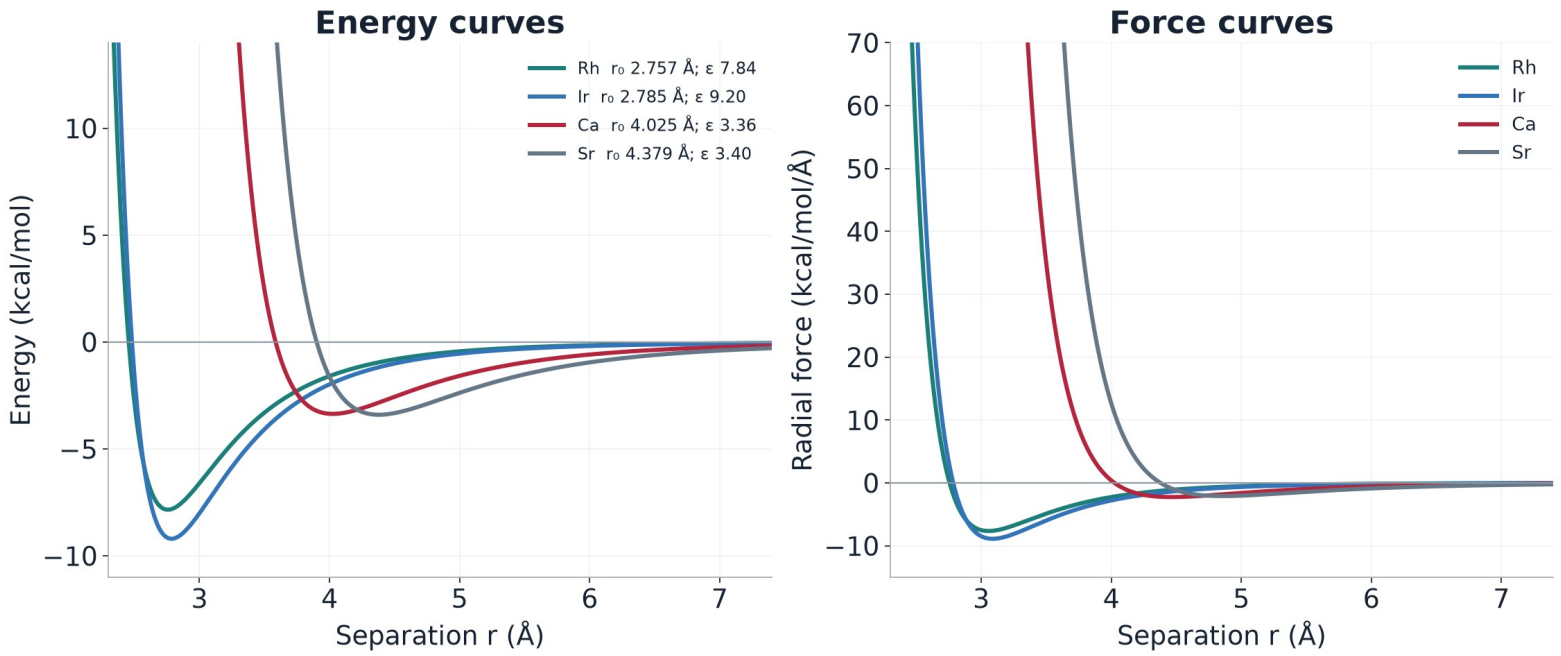
- At 3.000 Å, $U = 7.84[(2.757/3)^{12} - 2(2.757/3)^{6}] = -6.601$ kcal/mol.
- An energy below zero can still produce repulsion. The slope sets the force.

Kanhaiya et al., npj Comput. Mater. (2021) · Original calculation or illustration

Changing the metal changes the pair curve

Rhodium, iridium, calcium and strontium use one published parameter table.

- Rh and Ir have deeper, shorter wells in this set.
- Ca and Sr have wider, shallower wells.
- Each curve uses its own minimum length and well depth.



Kanhaiya et al. (2021), Table 1, p. 3. Curves are clipped at the upper plot boundary; no physical cutoff is applied.

A solid adds interactions with many neighbors

A pair well depth and a surface energy have different units.

Pair energy

ε describes one pair’s well depth in energy per mole of pairs.

Bulk cohesion

The model adds interactions across an extended atomic arrangement.

Surface work

Creating a surface changes contacts across an area.

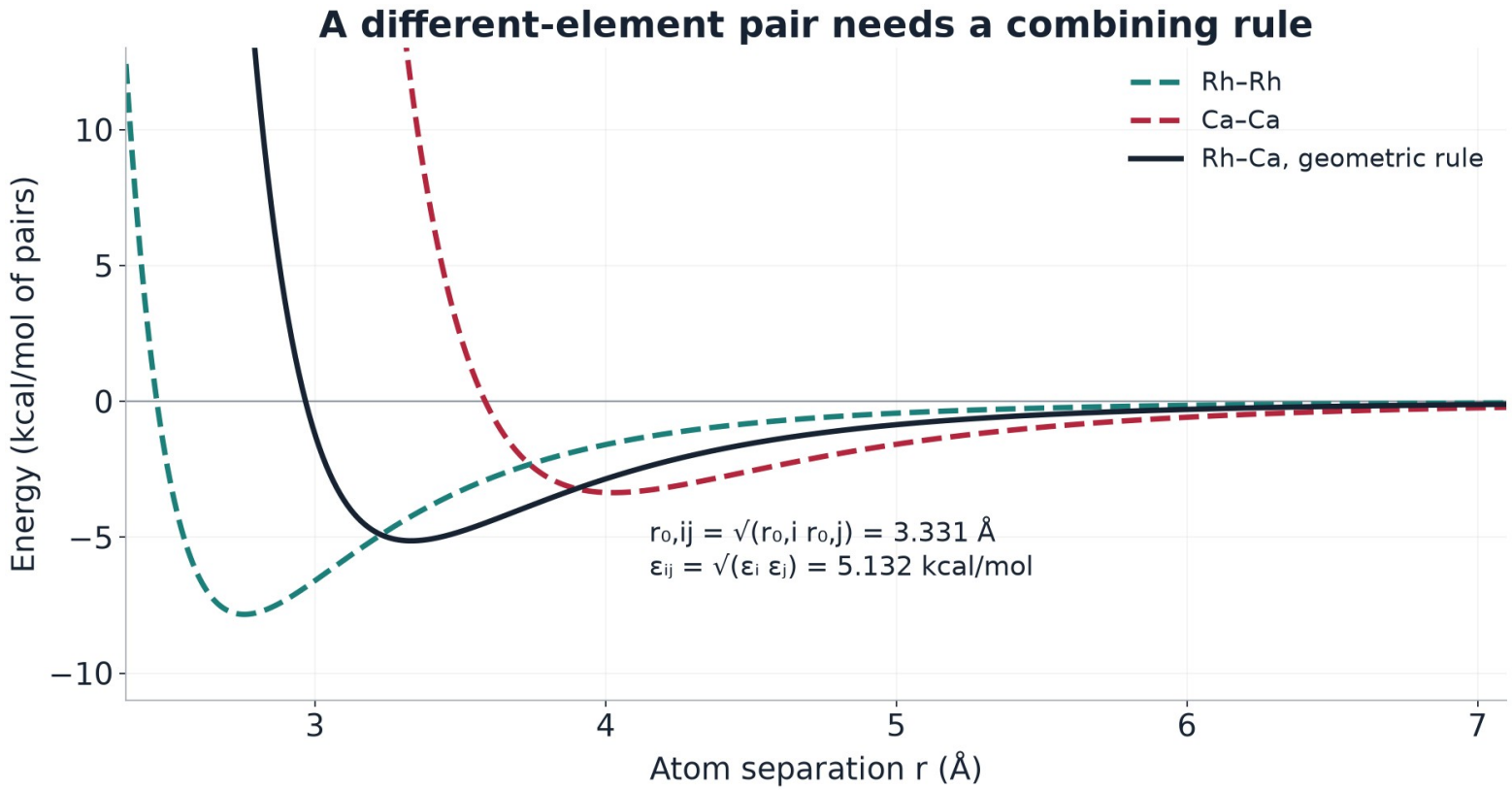
Connection

With fixed geometry and length parameters, the LJ energy sum scales with ε.

A mixed pair needs an explicit combining rule

This Rh–Ca example uses the CVFF geometric rule.

- $r_0,ij = \sqrt{(r_0,ii\ r_0,jj)} = 3.3312$ Å.
- $\varepsilon ij = \sqrt{(\varepsilon ii\ \varepsilon jj)} = 5.1325$ kcal/mol.
- At 4.000 Å, the mixed-pair energy is −2.8533 kcal/mol.

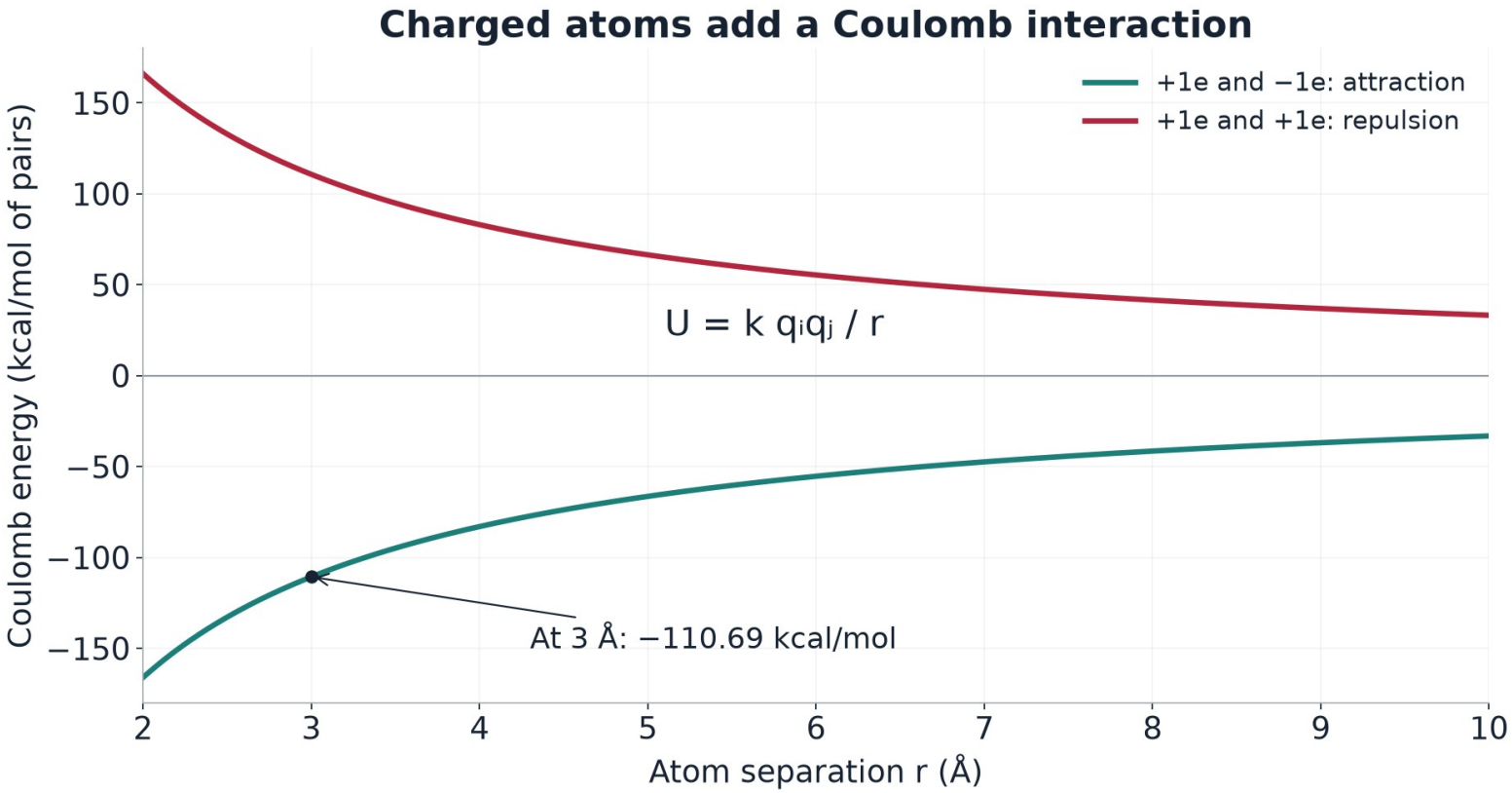


Explicit CVFF geometric convention. Arithmetic radii apply in other families. This is not validation of a Rh–Ca alloy.

Charges add electrical attraction or repulsion

Opposite model charges attract; equal signs repel.

- Coulomb energy is proportional to q_i q_j/r.
- For +1e and −1e at 3 Å in vacuum, U = −110.688 kcal/mol.
- Water, polarization and the ion model change the physical problem.



Illustrative point charges in vacuum, not Rh charges. Solvent, screening and periodic boundaries are absent.

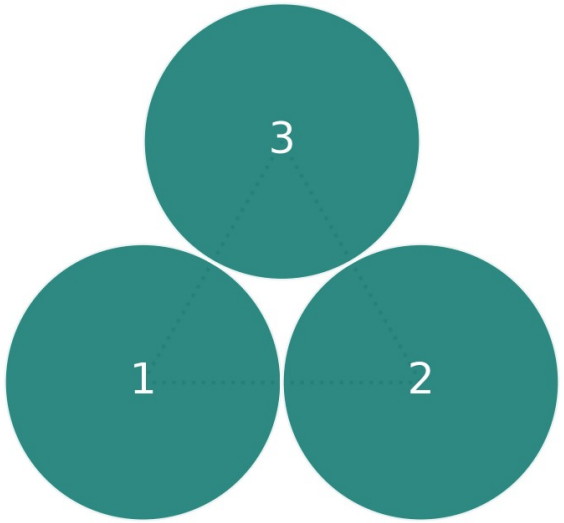
NIST, CODATA (2022) · Heinz et al., Langmuir (2013) · Original calculation or illustration

A small cluster counts each pair once

Three touching Rh atoms provide a transparent energy sum.

- The equilateral triangle has three distinct pairs.
- Each separation is r_0 = 2.757 Å; each pair contributes −7.84 kcal/mol.
- The total is −23.52 kcal/mol of clusters.

A tiny cluster adds its pair energies



3 atoms → 3 distinct pairs

Every separation is r_0 = 2.757 Å

Each pair: -7.84 kcal/mol

$U_{total} = U_{12} + U_{13} + U_{23}$
= -23.52 kcal/mol of clusters

Illustrative pair-minimum geometry. Sphere diameter = r_0. No cluster relaxation or stability prediction; not an fcc crystal.

Molecular mechanics makes repeated sampling practical

Simple interactions let us evaluate many configurations.

Motion

Follow atoms through many short time steps.

Interfaces

Include a surface, solvent and molecules in the same box.

Statistics

Estimate averages and fluctuations from sampled configurations.

Scope

Use a model tested for the chemistry and observables of interest.

Heinz et al., Langmuir (2013) · Kanhaiya et al., npj Comput. Mater. (2021)

Parameters connect equations to physical measurements

Each parameter has a physical role, with coupled effects.

Size and packing

Length parameters influence density and lattice spacing.

Cohesion

Energy parameters influence surface work and adhesion.

Polarity

Model charges influence electric interactions and adsorption.

Internal stiffness

Bond and angle terms influence geometry and vibrations.

Fitting and testing answer different questions

Reserve evidence that the fitting process cannot use.

Fit

Choose parameters against declared calibration targets.

Test

Evaluate another relevant observable after the choice is fixed.

If a test guides revision

Treat it as development evidence and find a fresh test.

Report

Keep reference uncertainty and model disagreement visible.

A parameter should have a defensible physical role

Interpretability helps explain a result and recognize a failure.

- Ask which measurements constrain the parameter.
- Check whether its size and sign make chemical sense.
- Change it deliberately and inspect affected observables.
- Recognize when the chosen equation cannot reproduce the behavior.

Heinz et al., Langmuir (2013) · Kanhaiya et al., npj Comput. Mater. (2021)

IFF connects inorganic surfaces with molecules

The Interface Force Field is a compatible parameter framework.

Inorganic part

Parameters and surface models for supported materials.

Molecular part

Compatible descriptions of water, polymers and biomolecules.

Interface

Cross interactions follow the chosen family’s equations and rules.

Evidence

Structural and energetic comparisons support a stated scope.

Heinz et al., Langmuir (2013)

Compatibility is an equation-level requirement

A familiar force-field name alone does not define the model.

Potential and units

Match the functional form and numerical units.

Length convention

Distinguish a pair minimum from a zero crossing.

Cross interactions

Preserve mixing rules, exclusions and scaling.

Readback

Check energies and forces from the files the engine consumes.

A silica surface changes its chemistry in water

pH and surface preparation affect which groups a molecule sees.

Surface groups

Silica can present neutral silanol groups and charged siloxide groups.

Solution

pH affects protonation; ions balance electrical charge.

Molecule

Positive peptide groups can pair with negative surface sites.

Model

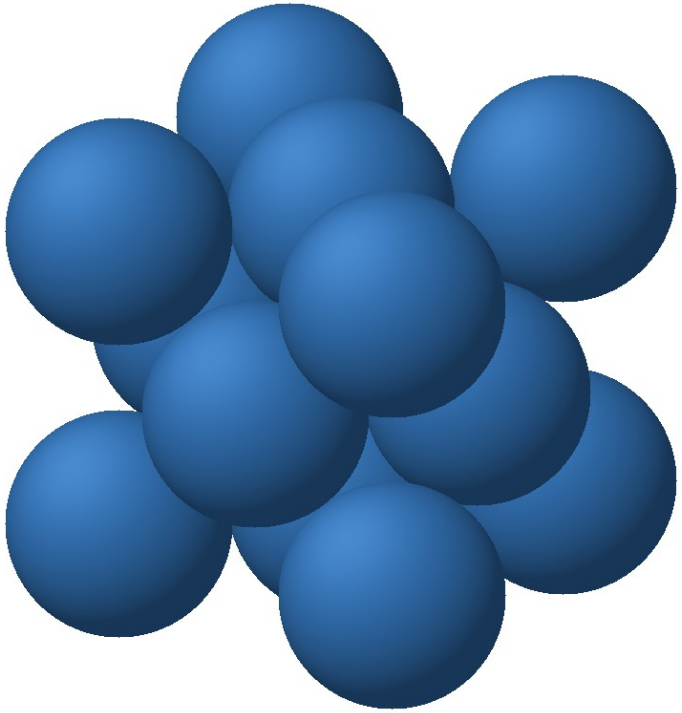
Choose surface chemistry to match the experimental sample.

Patwardhan et al., JACS (2012), doi:10.1021/ja211307u · Emami et al., Chem. Mater. (2014), doi:10.1021/cm500365c · Emami et al., correction (2016)

Rhodium begins with a repeating fcc cell

Face-centered cubic, or fcc, names the crystal arrangement.

- Atoms occupy the corners and face centers of a cube.
- Shared boundary atoms give four atoms per conventional cell.
- The reference cube has a = 3.8032 Å.

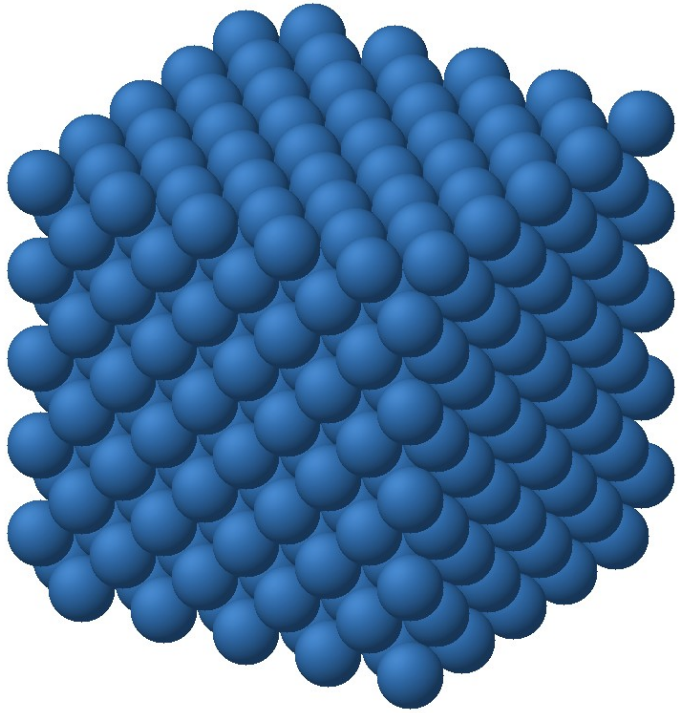


Conventional fcc cell. Shared corners and faces contribute four atoms per cell.

Repeat the cell to build a supercell

The boundary conditions define what lies beyond the box.

- A 5 × 5 × 5 array contains 500 atoms.
- Periodic boundaries repeat the box in space.
- Box size and periodic images can affect the result.

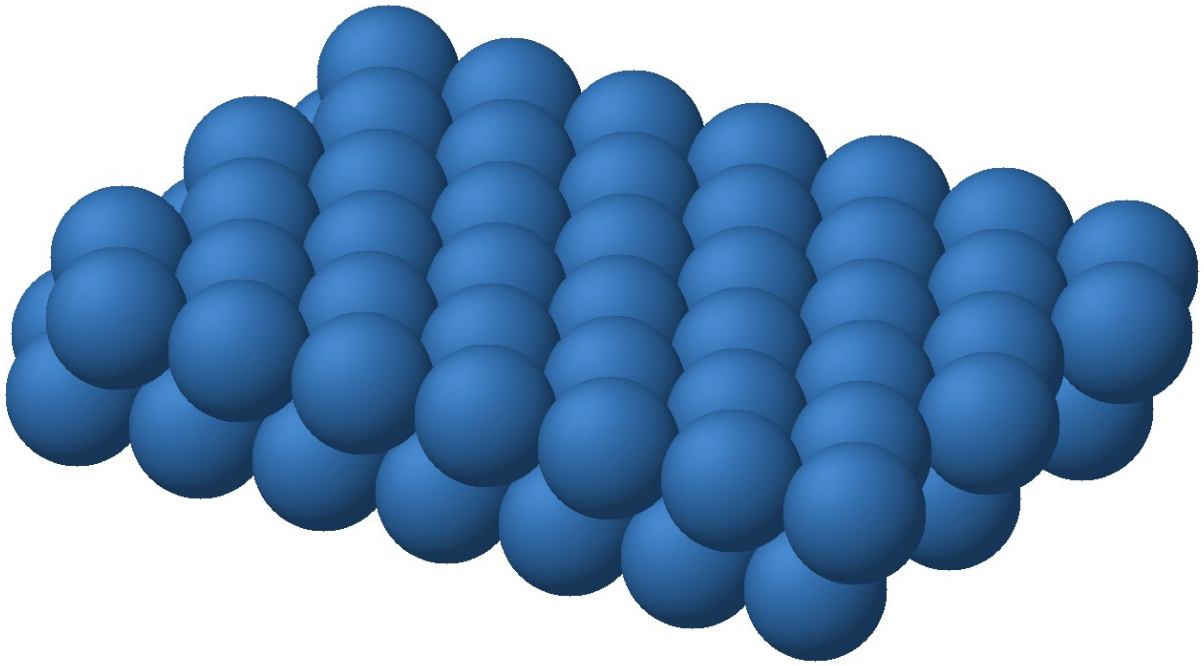


5 × 5 × 5 conventional fcc cells contain 500 atoms. Idealized geometry.

Expose a surface by cleaving the solid

The orientation determines which atomic layer is exposed.

- The label (111) identifies one crystallographic orientation.
- Separating a solid removes contacts across the cut.
- The surface atoms can relax after exposure.



Idealized close-packed surface. A slab exposes two faces.

Two new faces require a factor of two

Surface energy divides the cleavage work by the new area.

- $\gamma \approx (\langle E_{cleaved} \rangle - \langle E_{unified} \rangle) / (2A)$.
- A is the area of one face; cleaving makes two equivalent faces.
- Use matched atom counts and consistent simulation conditions.
- The energy-difference estimate needs a justified entropy approximation.

Excellent agreement can be a calibration result

These Rh measurements helped choose the metal parameters.

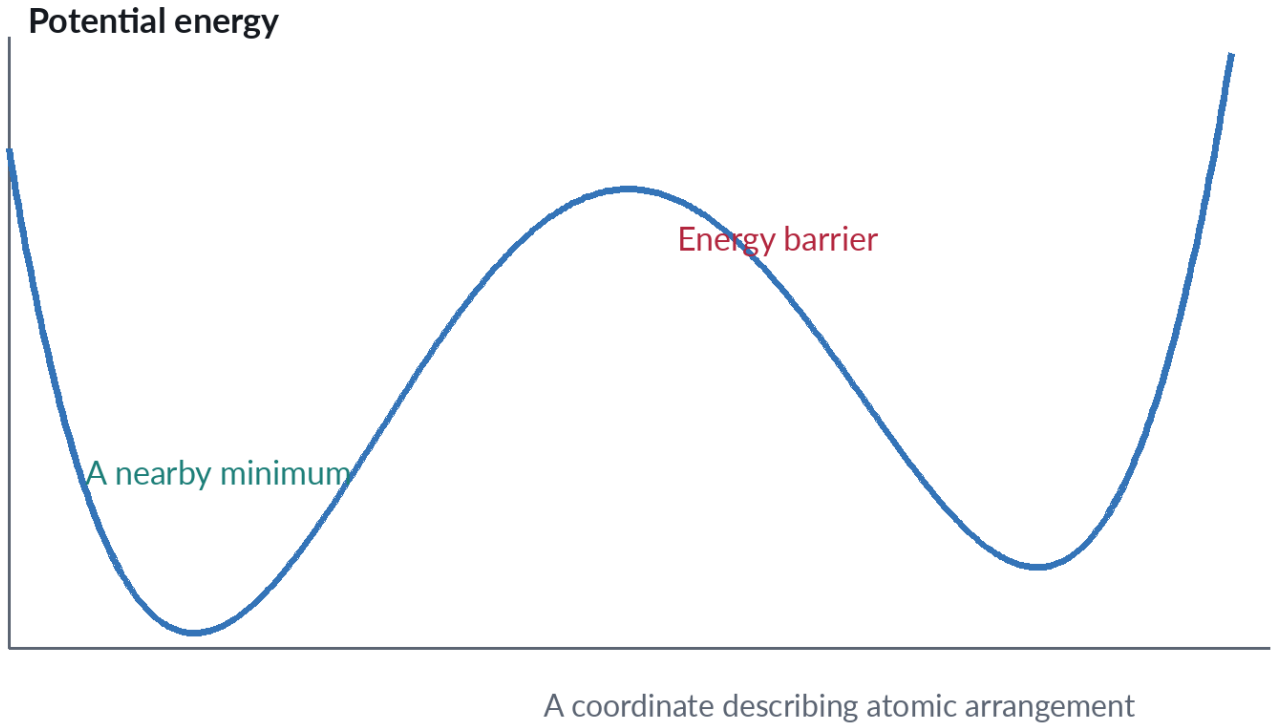
Calibration quantity	Reference	12-6 model	9-6 model
Five-cell length, Å	19.016	19.016	19.014
(111) surface energy, J/m²	2.64 ± 0.02	2.643	2.643

The surface reference can include polycrystalline samples. The paper explains its use as a (111) estimate.

Minimization finds a nearby low-energy arrangement

Move atoms downhill until the chosen stopping rule is met.

- Relax an initially strained geometry.
- The result depends on the starting configuration.
- A local minimum can coexist with lower minima elsewhere.

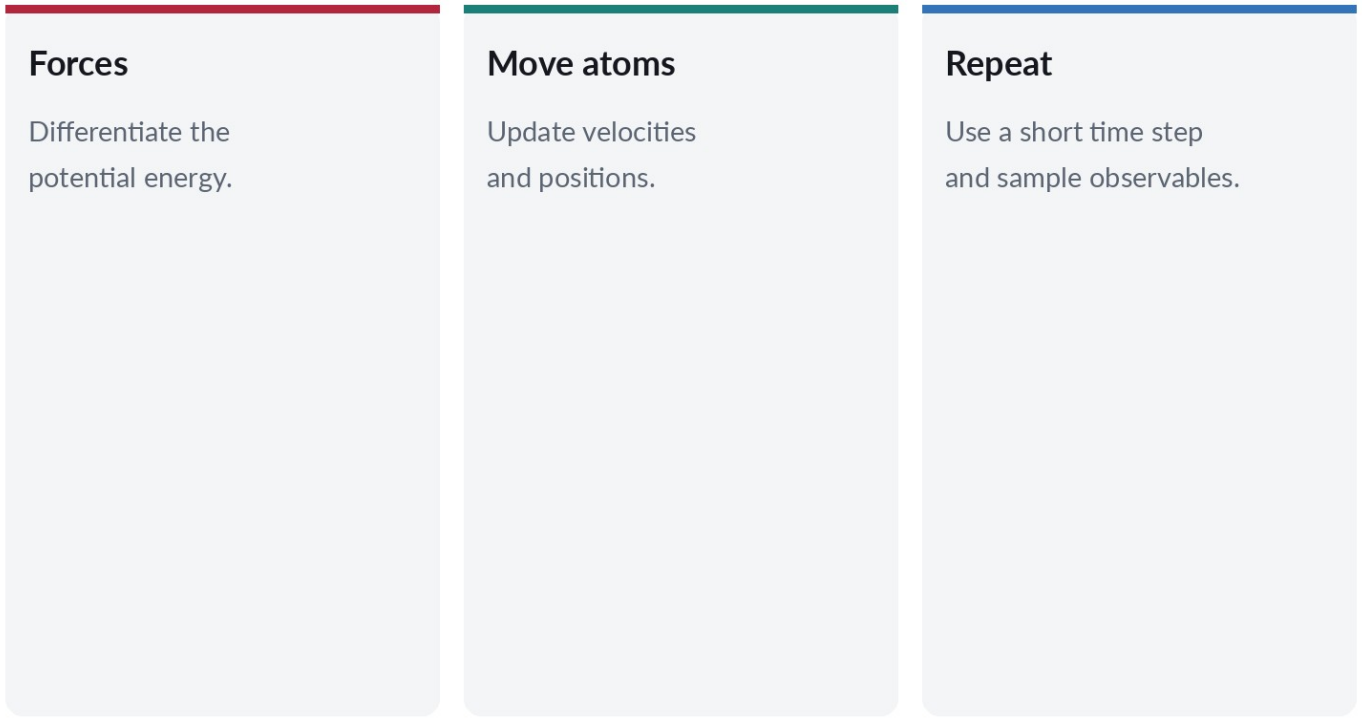


Illustrative landscape. A minimization can stop in a local minimum.

Molecular dynamics advances motion in short steps

Use forces and atomic masses to update positions and velocities.

- Calculate forces from the current configuration.
- Advance a short time step, then calculate again.
- The metal study uses a 1 fs time step, one quadrillionth of a second.



Molecular dynamics integrates an equation of motion. This is a workflow diagram.

A trajectory becomes evidence through an observable

Specify what is averaged and how uncertainty is assessed.

Density

Average the mass per sampled volume.

Surface energy

Compare matched energy averages and area.

Diffusion

Relate particle displacement to elapsed time.

Uncertainty

Check fluctuations, sampling duration and box size.

Rhodium mechanics exposes the pair model’s limit

Surface agreement does not establish mechanical accuracy.

Modulus, GPa	Reference selected	12-6	9-6
Bulk K	276	258	175
Young’s E	379	209	128
Shear G	147	77	47

The central-force pair framework constrains $C_{12}/C_{44} = 1$. Rh has larger errors in stretching and shear.

An alloy needs evidence beyond a mixing rule

AlNi illustrates why chemical assignment matters.

Composition

An alloy combines different elements in one material.

Charge model

The published AlNi model uses Al +0.39e and Ni −0.39e.

Constraint

The full structure must preserve overall electrical neutrality.

Evidence

Charge assignment uses alloy formation energetics and chemical reasoning.

Defects and polarization change the model question

A missing atom or a mobile electron cloud can require new terms.

Defect

Removing an atom changes its neighbors and the charge assignment.

AlNi evidence

Different charge-redistribution hypotheses change raw defect energies.

Polarization

A metal’s electron cloud responds to nearby charges.

Model choice

Compare a suitable pair model, many-body model or polarizable extension.

Liu et al., SI (2018) · Geada et al., Nat. Commun. (2018), doi:10.1038/s41467-018-03137-8 · Kanhaiya et al., npj Comput. Mater. (2021)

Real systems test different parts of a model

Each example names an experiment and a limit.

Platinum

Crystal-face preference and particle shape.

Silica

Peptide adsorption and water wetting.

Bone mineral

pH-dependent surfaces and hydration.

Cement and gold

Grain attraction and metal polarization.

REAL PUBLISHED SYSTEMS

Peptide chemistry steers platinum crystal shape

Can changing one amino acid alter which faces survive growth?

System

Peptides coat platinum nanocrystals growing in solution.

Model result

A phenyl ring creates binding contrast between (111) and (100) faces.

Experiment

Replacing L with F in TLT produced TFT and shifted cubes toward tetrahedra.

Limit

Shape is an indirect binding readout under the tested synthesis conditions.

Ruan et al., Nano Lett. (2013), doi:10.1021/nl400022g

A silica-binding peptide must fit the surface

Why can two changes to the same peptide have opposite effects?

System

Short peptides bind to silica nanoparticles in water.

Model result

Charge pairing, hydrogen bonds and molecular flexibility control contacts.

Experiment

Two histidine-to-alanine mutations gave opposite adsorption trends.

Limit

The modeled surface chemistry must represent the measured particles.

Patwardhan et al., JACS (2012), doi:10.1021/ja211307u

Surface groups change how water wets silica

Can a model reproduce the heat released when dry silica meets water?

System

Silica surfaces with different exposed oxygen and hydroxyl groups.

Model result

Water affinity changes with the surface groups.

Experiment

Immersion heat was 160 ± 5 mJ/m²; the TIP3P simulation gave 160 ± 2.

Limit

Hydration energy helped choose the parameters, so this is a calibration comparison.

Emami et al., Chem. Mater. (2014), doi:10.1021/cm500365c · Emami et al., correction (2016)

Bone mineral presents a pH-dependent surface

Which surface does water or a biomolecule actually encounter?

System

Hydroxyapatite, a major mineral in bone and teeth.

Model result

Protonation changes surface composition, hydration and facet stability.

Experiment

Facet-averaged immersion energies were compared with calorimetry.

Limit

Surface preparation and hydration depth complicate a one-to-one comparison.

Lin & Heinz, J. Phys. Chem. C (2016), doi:10.1021/acs.jpcc.5b12504

Grinding aids reduce attraction between cement grains

Why can a little organic additive help freshly broken grains stay apart?

System

Tricalcium silicate surfaces with organic amines and alcohols.

Model result

Adsorbed molecules separate surfaces and weaken electrical attraction.

Experiment

Predicted trends correlate with measured grinding-aid efficiency.

Limit

Ideal face-to-face attraction is one part of a working cement mill.

Mishra et al., J. Phys. Chem. C (2013), doi:10.1021/jp312815g

Gold’s electrons respond to nearby charges

When does a fixed metal pair model miss an electrical response?

System

Gold in contact with water, ions and charged peptides.

Model result

A polarizable extension responds to nearby charges and alters ion binding.

Experiment

Water-monolayer adsorption was −13.0 ± 0.1 kcal/mol on modeled Au(111); references were −13 and −15.5.

Limit

The experimental references are polycrystalline; ion profiles are model predictions.

Geada et al., Nat. Commun. (2018), doi:10.1038/s41467-018-03137-8

Choose the observable before judging the result

The experimental question determines the strength of the claim.

Shape

Morphology can support a binding mechanism while growth also matters.

Adsorption

Match molecule, surface state and solution.

Surface energy

Keep reference state, area and entropy assumptions explicit.

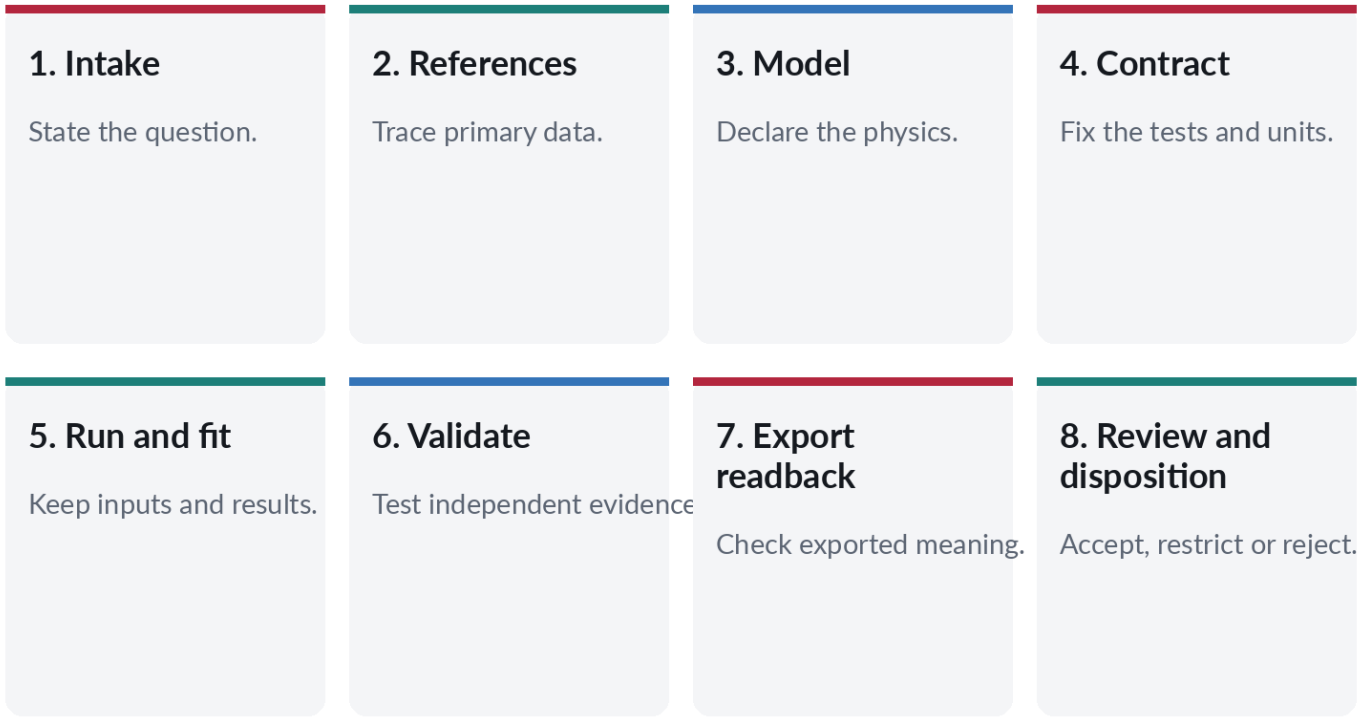
Prediction

State whether the comparison was used to fit the parameters.

The IFF Agent carries evidence through eight steps

The workflow connects the material question to a reviewed result.

- Scientific choices remain linked to references.
- The model and its exported interpretation stay inspectable.
- Review can qualify, restrict or withhold a result.



Workflow repeatability does not by itself establish scientific accuracy.

Define the question before running the fit

The first four steps establish what the result must mean.

1. Intake

Specify material, phase, conditions and intended observable.

2. References

Select primary evidence and keep its uncertainty.

3. Model

Assign chemical types, charges and a force-field family.

4. Contract

Declare calibration targets, reserved tests and acceptance limits.

Run, test, read back, then review

The remaining four steps preserve the meaning of the result.

5. Run and fit

Calculate with source-grounded structures and declared settings.

6. Validate

Check reserved observables and convergence.

7. Export readback

Compare the actual files consumed by each engine.

8. Review and disposition

Record a qualified result and its supported scope.

A repeatable answer can still be inaccurate

Repeatability checks execution; accuracy needs an external reference.

Repeatability

Reuse the same inputs and obtain consistent outputs.

Accuracy

Compare the relevant observable against suitable evidence.

Model error

A stable, converged calculation can disagree with experiment.

Example

The Rh pair model repeats its limited mechanical response.

Use DFT when electron rearrangement matters

Electronic structure can be the central unknown.

- Investigate bonding changes, charge transfer or local electronic states.
- Choose and test the density-functional approximation.
- Check system size, sampling and reference dependence.
- Use suitable experiments or higher-level calculations for comparison.

Use classical IFF for supported interface sampling

A validated effective model can describe many configurations.

- Represent the material's chemistry and surface state.
- Use compatible molecular and inorganic parameters.
- Sample long enough for the intended observable.
- Test transfer to a changed composition or environment.

Heinz et al., Langmuir (2013) · Kanhaiya et al., npj Comput. Mater. (2021) · Geada et al., Nat. Commun. (2018), doi:10.1038/s41467-018-03137-8

Use a learned potential within its supported data domain

A model's training coverage shapes the questions it can answer.

- Train against suitable reference energies and forces.
- Include relevant structures, compositions and conditions.
- Test on evidence that did not guide development.
- Examine uncertainty when entering a new environment.

Batatia et al., MACE (2022) · Wang et al., Nature (2024)

Compare methods against the same physical question

The useful choice depends on detail, sampling and evidence.

Method	Useful physical detail	What enables scale	What needs scrutiny
DFT	Electronic structure	Approximate electronic theory	Functional and sampling
Classical IFF	Effective atom interactions	Simple evaluated equations	Parameters and chemistry
Learned potential	Reference-data energy landscape	Trained approximation	Coverage and extrapolation

Hohenberg & Kohn, Phys. Rev. (1964) · Heinz et al., Langmuir (2013) · Batatia et al., MACE (2022) · Wang et al., Nature (2024)

What these figures establish

Each visual has a specific evidential role.

Structures

Idealized space-filling models explain geometry.

Pair plots

Analytic evaluations use published parameters and stated equations.

Motion

Illustrative movement explains an operation or force direction.

Material comparisons

Published results are cited and were not rerun here.

Original calculation or illustration · Kanhaiya et al., npj Comput. Mater. (2021)

Ask what would change your confidence

A useful review can name both supporting evidence and failure conditions.

Chemistry

Does the structure match the phase, surface and solution?

Reference

Is the observable and its uncertainty suitable?

Sampling

Do time, size and numerical choices leave the result stable?

Scope

Which new observation would require a different model?

Heinz et al., Langmuir (2013) · Kanhaiya et al., npj Comput. Mater. (2021) · Patwardhan et al., JACS (2012), doi:10.1021/ja211307u · Lin & Heinz, J. Phys. Chem. C (2016), doi:10.1021/acs.jpcc.5b12504 · IFF Agent workflow

IFF makes an interface a testable atomic model

Its value comes from chemistry, compatibility and evidence together.

- Connect a solid with water, polymers or biomolecules.
- Give the parameters a physical interpretation.
- Check structural and energetic properties against stated measurements.
- Carry the result with conditions, uncertainty and limits.

Heinz et al., Langmuir (2013)

Some questions require a different model

Change the method when the missing physics controls the answer.

Reactions

Bond formation and breaking require suitable reactive or electronic treatment.

Changing charges

Polarization and charge transfer need appropriate terms.

New defects or phases

Reassess chemical assignments and evidence.

Unsupported interfaces

Obtain relevant parameters and independent tests.

Heinz et al., Langmuir (2013) · Kanhaiya et al., npj Comput. Mater. (2021) · Geada et al., Nat. Commun. (2018), doi:10.1038/s41467-018-03137-8 · Liu et al., SI (2018)

Read the primary evidence

The full bibliography and source notes travel with the deck.

- Foundation | Heinz et al. 2013, IFF framework.
- Worked metal | Kanhaiya et al. 2021, paper and supplementary methods.
- Applications | Ruan 2013; Patwardhan 2012; Emami 2014; Lin 2016; Mishra 2013; Geada 2018.
- Methods | Hohenberg and Kohn 1964; Batatia et al. 2022; Wang et al. 2024.

Questions

Which observation would make us change the model?

- What is the physical question?
- Which measurement would test the answer?
- Which missing effect could change the conclusion?

Equations and units

Use one declared convention throughout a calculation.

- Utotal = Ubonds + Uangles + Utorsions + UCoulomb + UvdW.
- U12-6 = $\varepsilon[(r_0/r)^{12} - 2(r_0/r)^6]$.
- $F_r = (12\varepsilon/r)[(r_0/r)^{12} - (r_0/r)^6]$.
- UCoulomb = kq_i q_j/r. Energy is kcal/mol; length is Å; force is kcal/(mol Å).

APPENDIX

The paper’s sigma is a minimum-length parameter

Read the equation before transferring a coefficient.

- Paper form uses σpaper = r_0 in $\varepsilon[(r_0/r)^{12} - 2(r_0/r)^6]$.
- Textbook and lj/cut form uses $4\varepsilon[(\sigma zero/r)^{12} - (\sigma zero/r)^6]$.
- The conversion is σzero = r_0 / 2^(1/6).
- Rh gives 2.757 Å / 2^(1/6) = 2.4562077659 Å.

Kanhaiya et al., npj Comput. Mater. (2021) · LAMMPS, lj/cut documentation · Original calculation or illustration

APPENDIX

The 9-6 and 12-6 wells are different functions

Each form has its own fitted coefficients and mixing rules.

12-6

$U = \varepsilon[(r_0/r)^{12} - 2(r_0/r)^6]$. Rh uses $r_0 = 2.757$ Å and $\varepsilon = 7.84$ kcal/mol.

9-6

$U = \varepsilon[2(r_0/r)^9 - 3(r_0/r)^6]$. Rh uses $r_0 = 2.807$ Å and $\varepsilon = 6.38$ kcal/mol.

At the minimum

Both written forms give $U(r_0) = -\varepsilon$ and zero radial force.

Transfer

Changing the radius alone cannot convert one function into the other.

Heinz et al., Langmuir (2013) · Kanhaiya et al., npj Comput. Mater. (2021) · Original calculation or illustration

APPENDIX

Parameters and worked-number checks

Published coefficients feed independently checked analytic calculations.

Metal	r_0, Å	ε, kcal/mol	σzero, Å
Rh	2.757	7.84	2.4562
Ir	2.785	9.20	2.4812
Ca (α)	4.025	3.36	3.5859
Sr (α)	4.379	3.40	3.9012

Derived checks include mixed Rh–Ca ε = 5.1325, vacuum charge energy −110.688, and triangle energy −23.52 kcal/mol.

APPENDIX

Rhodium evidence and uncertainty

Calibration, prediction and reference uncertainty remain separate.

Quantity	Reference	12-6	9-6
5a, Å	19.016	19.016	19.014
γ111, J/m²	2.64 ± 0.02	2.643	2.643
K, GPa	276	258	175
Poisson ratio	0.26	0.365	0.378

- Other K references are 270, 271 and 269 GPa. The selected reference here is 276 GPa.
- The paper reports $C_{12}/C_{44} = 1$ for its central-force pair framework.

Kanhaiya et al., npj Comput. Mater. (2021) · Kanhaiya et al., SI (2021) · Liu et al., SI (2018)

APPENDIX

References and visual provenance

All diagrams and plots are original; papers support the scientific facts.

- IFF | Heinz et al. Langmuir 2013. DOI 10.1021/la3038846.
- Metals | Kanhaiya et al. 2021. DOI 10.1038/s41524-020-00478-1; correction 10.1038/s41524-021-00576-8.
- Applications | Ruan, Patwardhan, Emami, Lin, Mishra and Geada papers are listed in the citation manifest.
- Rebuild | Source equations, parameters, notes and licenses accompany the deck.